RELATIVE FROBENIUS IS AN ITERATED ALPHA_P-COVER OF DEGREE P^D

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file crystalline.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original lemma 07Q9: sources/stacks-chow/tags.txt, source lines 9824–9824. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07QA: sources/stacks-chow/tags.txt, source lines 9825–9825. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 054L: sources/stacks-chow/tags.txt, source lines 6644–6644. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0EBS: sources/stacks-chow/tags.txt, source lines 17932–17932. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In the situation above, assume that $X \to S_0$ is smooth of relative dimension d. Then F_{X/S_0} is an iterated α_p-cover of degree p^d. Hence Lemmas 07Q9 and 07QA apply to this situation. In particular, for any crystal in quasi-coherent modules $\mathcal{G}$ on $Cris(X^{(1)}/S)$ the map*

$$F^*_{X/S_0} : H^i(Cris(X^{(1)}/S), \mathcal{G}) \longrightarrow H^i(Cris(X/S), F^*_{X/S_0,cris}\mathcal{G})$$

has kernel and cokernel annihilated by $p^{d(i+1)}$.

Proof. It suffices to prove the first statement. To see this we may assume that X is étale over $\mathbf{A}^d_{S_0}$, see Morphisms, Lemma 054L. Denote $\varphi : X \to \mathbf{A}^d_{S_0}$ this étale morphism. In this case the relative Frobenius of X/S_0 fits into a diagram

$$\begin{array}{ccc} X & \longrightarrow & X^{(1)} \\ \downarrow & & \downarrow \\ \mathbf{A}^d_{S_0} & \longrightarrow & \mathbf{A}^d_{S_0} \end{array}$$

where the lower horizontal arrow is the relative frobenius morphism of $\mathbf{A}^d_{S_0}$ over S_0. This is the morphism which raises all the coordinates to the pth power, hence it is an iterated α_p-cover. The proof is finished by observing that the diagram is a fibre square, see Étale Morphisms, Lemma 0EBS. □

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